Youngest Common Ancestor

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The problem

- Nodes have only an `ancestor` pointer; find the deepest common ancestor of two descendants.
- Equalise depths, then walk up together.

Intuition

- Bring both descendants to the same depth, then climb together until they meet.

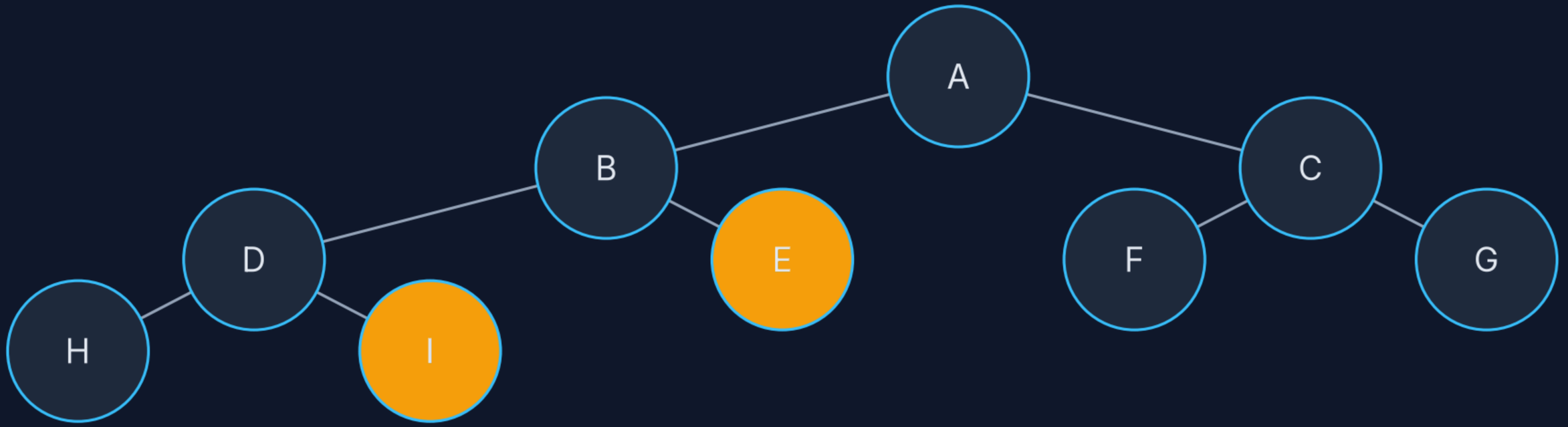
Brute force vs optimal

- Store all ancestors of one node in a set, walk up the other → O(d) space
- Equalise depths, then climb together → O(d) time, O(1) space
- D = depth of the deeper node

Approach

- 1. Compute both depths
- 2. Lift the deeper one
- 3. Move both up until equal

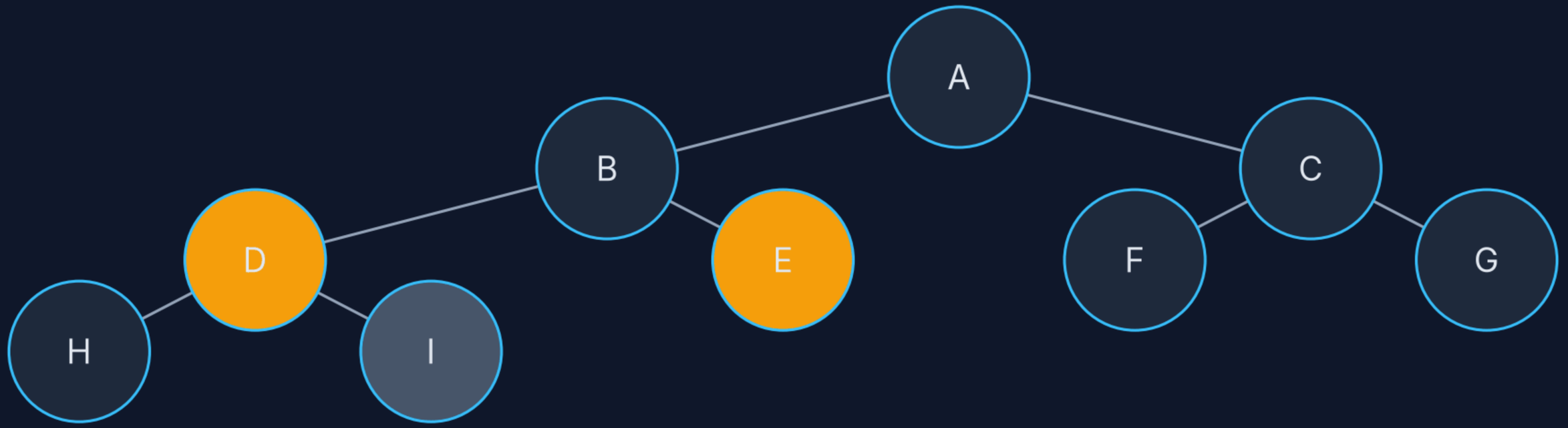
Worked example




```
depth(E) = 2, depth(I) = 3
```

Descendants E and I; depths 2 and 3

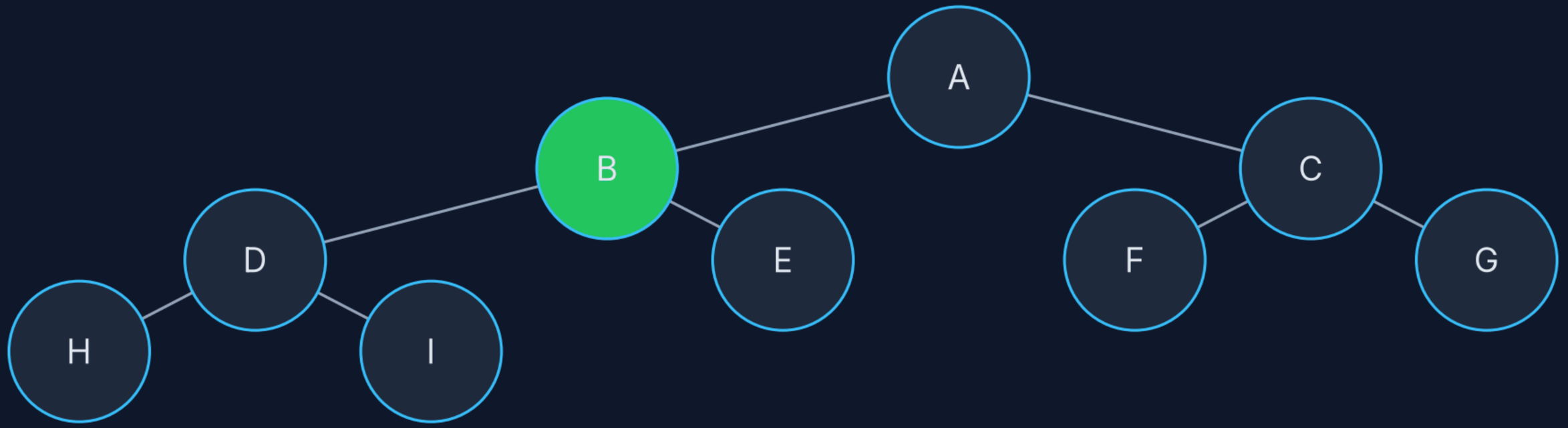
Worked example




```
lift I one level → D
```

Lift the deeper one: I → D

Worked example




```
E → B, D → B: equal
```

Move both up: E → B, D → B → answer B

Java solution

```java
public class YoungestCommonAncestor {
    public static class AncestralTree {
        public final char name;
        public AncestralTree ancestor;
        public AncestralTree(char name) { this.name = name; }
    }

    public static AncestralTree getYoungestCommonAncestor(AncestralTree top, AncestralTree a, AncestralTree b) {
        int da = depth(a, top), db = depth(b, top);
        while (da > db) { a = a.ancestor; da--; }
        while (db > da) { b = b.ancestor; db--; }
        while (a != b) { a = a.ancestor; b = b.ancestor; }
        return a;
    }

    private static int depth(AncestralTree n, AncestralTree top) {
        int d = 0;
        while (n != top) { n = n.ancestor; d++; }
        return d;
    }
}
```

Complexity

- Time: O(d)
- Space: O(1)

Edge cases & pitfalls

- One node is the ancestor of the other
- Both nodes are the same
- The answer may be the top ancestor

Interview tips & follow-ups

- Follow-up: binary tree without parent pointers (LCA)
- Follow-up: many queries → binary lifting

Recap

- Bring both descendants to the same depth, then climb together until they meet.
- Complexity: O(d)